

Assignment 2:

Object-Oriented Software Engineering

Objective:

Based on the **project definition submitted in Assignment 1**, prepare a detailed software project estimation using different Object-Oriented estimation techniques and perform risk analysis for the proposed system.

Tasks

1. Project Estimation

- a. Apply the **Lorenz and Kidd Class Point** method to estimate the size and effort of your project based on the object-oriented classes identified in Assignment 1.
- b. Apply the **Use Case Point (UCP)** method using the actors, use cases, technical factors and environmental factors identified for your project.
- c. Apply the **Class Point** method using the key and supporting classes identified for your project.
- d. Apply **Object-Oriented Function Point (OOF) Analysis** based on the functional requirements, inputs, outputs, inquiries, logical files and external interfaces of your system.
- e. Convert the estimated size obtained from each method into **effort/person-hours/person-months**, using clearly stated and justified productivity assumptions.

2. Comparison and Analysis

- a. Compare and contrast the estimation results obtained using **Lorenz & Kidd, Class Point, Use Case Point and OOF Analysis**.
- b. Present the results in a comparative table.
- c. Identify which estimation method you consider most appropriate for your project and justify your choice.

3. Risk Analysis

- a. Identify at least **10 major risks** associated with your project.
- b. Categorize the risks as technical, requirements, project management, operational, resource, security, or other relevant categories.
- c. Assign **Probability** and **Impact** values to each risk.
- d. Prepare a **Risk Exposure Matrix** and prioritize the identified risks.

4. RMMM Plan

- a. Prepare an **RMMM (Risk Mitigation, Monitoring and Management) plan** for the major/high-exposure risks.

Submission Requirements

- Use the **same project defined in Assignment 1**.
- Clearly state all assumptions used for estimation.
- Show **all formulas, intermediate calculations and final results**.
- Present the three estimation results in a **comparative table**.
- Include a **Risk Exposure Matrix** and **RMMM table**.
- The analysis should demonstrate your own understanding and application of the estimation techniques rather than only presenting theoretical definitions.

Deliverable: Submit a structured report containing the estimation calculations, comparative analysis, risk exposure matrix and RMMM plan.

Upload your assignment with the title “OOSE A2_Roll No_Name.pdf”